SINKGRAB · DESIGN DECISION RECORD

SinkGrab design for construction

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DOCUMENT CONTROL

Document ID	SKG-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/decisions/0002-design-for-construction.md @ 780e4b8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Constructability review and changes that make the design buildable, decided under Amish's pre-approvals of 2026-10-03

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted

Context

STANDARDS section 18 asks for every part to be makeable by its stated process and to fit and fasten to its neighbours (Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible"). The TRL 2 concept (SKG-PRC-001 v0.2, SKG-DDR-001) was modelled part by part in `cad/src/model.py`: how each part is made, how it joins each neighbour, and build123d checks for overlapping parts, for parts that do not touch what holds them, and for the folded scraper passing the smallest ring. The checks now report no overlaps and no floating parts. Amish pre-approved every recommendation on 2026-10-03 ("start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost.") and extended it the same day to this batch ("Proceed with the remaining 15 scaffolds"). No change alters what SinkGrab does or its pitch; the safety-related changes all take the conservative side.

Options considered

For each problem the simplest physically sound fix was chosen; alternatives are noted in Table 1.

Decision

Table 1. Changes made for construction.

#	Part	Problem found	Change	Why
1	Capstan position	A 69 kg capstan cannot hang on the tripod's aluminium leg	Ground capstan 3.0 m from a lead sheave; drawbar of two 48.3 mm halves to a cradle under the leg A foot	Fleet angle 1.9 degrees; the pull closes inside the kit
2	Head sheave	The HatchSide sheave is grooved for 6 mm wire	A bought 150 mm sheave grooved for 8 mm fibre rope fits the same cheeks and axle	Groove fit; 16 rope diameters
3	Capstan frame	First model 32 kg in 40 x 40 x 2.6 tube with a solid anchor bar	40 x 40 x 2 tube, 8 mm pads, anchor bar as tube: 22.9 kg	R9
4	Drum	8 mm solid flanges and webs	6 mm flanges and brake web with lightening holes: 18.3 kg	R9
5	Pawl	A pawl pivoted ahead of the nose ran its body through the tooth tips	Pivot 150 mm from the axis at 75 degrees, behind the nose; bracket hung from the front brace	Clears the teeth; holds the paying-out direction
6	Brake	Not detailed	Lined band 270 degrees round a 250 mm brake drum; anchor post, link, 650 mm lever with a 6 kg weight; anchor and link moved apart so they cannot touch	Brake on when let go; 314 N m
7	Stakes and guard	Stake heads clashed with the braces; guard tabs inside the guard wall	Stake tubes 200 mm either side of the axis; frame tabs reach the guard face	Fit
8	Grab mass	First model 72 kg, too close to the link release with a full bite	4 mm skin, 6 mm end plates, 4 mm top cover, hollow 6 mm head box, 30 x 10 tie rods: 47.9 kg	Link margin 1.19
9	Grab hinge	Both shells' end plates in one plane at the pin	Shell B's end plates outside shell A's; each shell's tie rods outside its own end plates; short pins at the lugs	Interleaved hinge, no clash
10	Shear link	Not detailed	Two 6 mm plates on a 16 mm pin through the dead-end lug; calibrated pin of about 1.9 mm below	R7; opens the grab if it snags
11	Scraper arms	Arms fixed open would not pass the bore	Arms pinned to the hub, opened by struts from a sliding sleeve and foot; a 300 mm lift folds the toes to 364 mm radius	R2
12	Scraper pole	Pole sections were not defined	42.4 x 2.6 tube in 2.0 m sections with 36 mm spigots and M12 cross bolts; T-bar socket and eye	R9 (fits a small pickup)
13	Spoil and cover	A chute over the mouth could not also cover the shaft	Well-head frame in two halves on the collar with two mesh doors, a 64 mm pole hole and four datum brackets	Covers the shaft; pole guide; tilt datum
14	Ballast	A frame on the top ring needed placing by hand	Eight 20.6 kg U-saddles straddling the ring wall, each on a 6 mm line	Placed from the surface
15	Clearances		Holes 0.25 to 0.5 mm over the pin	

#	Part	Problem found	Change	Why
		Pins in 1 mm oversize holes left parts unconnected in the checks		Each part rests on what holds it

Consequences

- `cad/src/model.py` holds the constructable design; STEP and STL, SKG-DWG-001 and 002 (Rev P2), the concept media and the build plan pictures are generated from it.
- SKG-CAL-001 is run on it; the parts cost is USD 2,506, USD 1,494 under the value-engineering target.
- `design_state: constructable` is set in `project.yaml`.